Human-Supervised Concept Bottlenecks for Interpretable Bangla Text Classification

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Abstract

Concept Bottleneck Models (CBMs) explain predictions through human-readable concepts, yet textual CBMs usually obtain concept scores automatically, and those scores are rarely checked against dense human judgement. We examine this in Bangla sarcasm detection, where meaning depends on pragmatic context rather than wording alone. From Ben-Sarc, we discover 43 concepts without a predefined ontology and have three annotators independently score 10,162 comments against every concept. The resulting 436,966 instance–concept judgments (mean pairwise Cohen's $\kappa = 0.83$) are, to our knowledge, the largest dense human concept annotation for a textual CBM, and the first for Bangla. Across nine encoders, human-supervised CBMs achieve higher macro-F1 than embedding-similarity-supervised CBMs on every encoder (by roughly 6–12 points), while retaining 95–99% of direct-classification accuracy on eight. The same matrix exposes a gap that task accuracy hides: GPT-5.4 and GPT-5.6 reach about 79% polarity accuracy, above every model we train, yet agree with human concept judgements only modestly (macro-F1 0.41 and 0.36; mean per-concept κ 0.39 and 0.34, versus 0.83 among annotators). GPT-4o is weaker on both (65.9% accuracy, macro-F1 0.27). Accurate predictions do not guarantee concepts a human would recognise.

1 Introduction

Consider a comment posted minutes after a cricket team is bowled out for 80: "What a performance!" In isolation, it reads as praise; in context, it may be sarcastic. The words alone do not reveal the speaker's intent. A conventional classifier can learn such cases from task labels, but its hidden representation does not expose the human-interpretable evidence behind the prediction.

Concept Bottleneck Models (CBMs; Koh et al., 2020) address this limitation by predicting human-readable concepts before the final label. The concept layer makes intermediate evidence inspectable and, in principle, intervenable. CBMs have evolved from human-defined concepts in vision (Koh et al., 2020) toward approaches that reduce exhaustive annotation (Oikarinen et al., 2023), and have recently been applied to text through Text Bottleneck Models (Ludan et al., 2024), C^3M (Tan et al., 2025), CB-LLM (Sun et al., 2025), CT-CBM (Bhan et al., 2025), and SCBM (Labadie-Tamayo et al., 2025).

This raises a central question: *how closely do automatically obtained concept scores correspond to concepts humans assign to text?* The question is particularly important for pragmatic phenomena, where semantic similarity may not capture communicative intent. Yet, to our knowledge, textual CBMs have not been evaluated with a dense human matrix in which every instance is scored against every concept.

We study this question in Bangla sarcasm detection, where interpretation depends on irony, contextual contrast, and cultural reference (Joshi et al., 2017). Using Ben-Sarc (Lora et al., 2025), we construct a 43-concept vocabulary without a predefined ontology and obtain dense human concept supervision for 10,162 comments. Three annotators independently score every comment against every concept, producing 436,966 instance–concept judgments with mean pairwise Cohen's $\kappa = 0.83$. To our knowledge, this is the largest dense human concept annotation reported for a textual CBM by instance–concept judgments, and the first such resource for Bangla.

We evaluate the resource across nine encoders using 27 independently trained runs: direct classifiers, human-supervised CBMs, and similarity-supervised CBMs. Human supervision yields 5.8–11.9 macro-F1 points higher than embedding-similarity supervision across the nine encoders. We additionally use the human matrix to examine GPT-4o, GPT-5.4, and GPT-5.6 as automatic concept

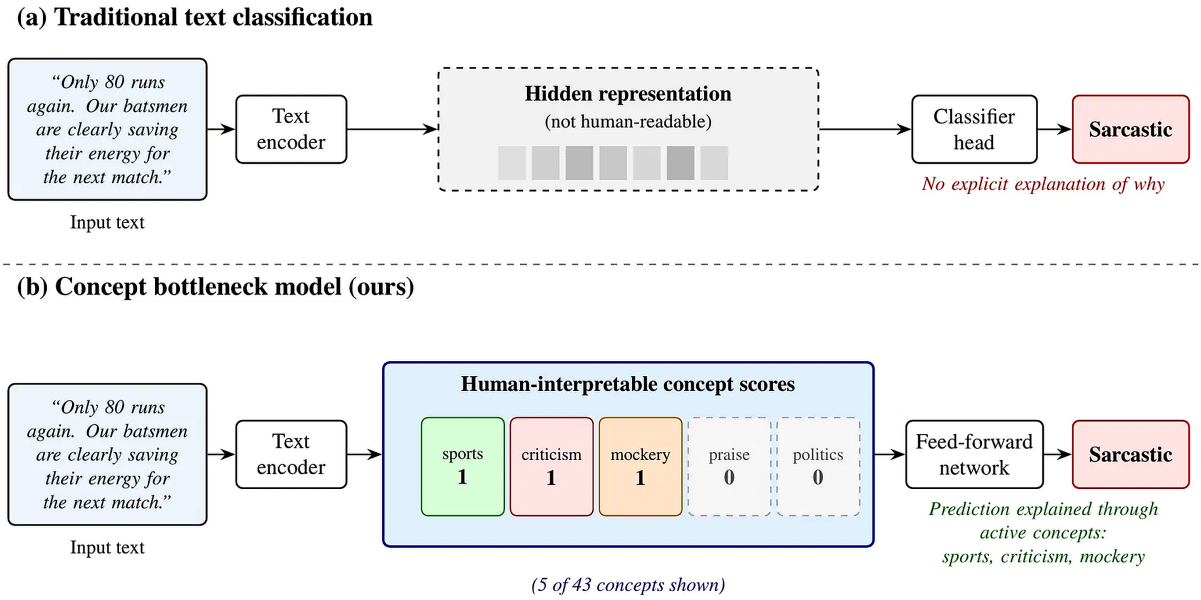


Figure 1: Traditional text classification versus our Concept Bottleneck Model (CBM). A conventional classifier maps text directly to a label through a hidden representation, whereas our CBM routes the prediction through human-interpretable concept scores, making the active concepts behind the decision inspectable.

scorers, separating task-level prediction from agreement with human concepts.

Our contributions are:

- **A dense human concept resource:** 436,966 instance–concept judgments over 43 concepts, providing both CBM supervision and a human reference for concept scorers.
- **A data-driven Bangla CBM:** a 43-concept vocabulary constructed without a predefined ontology and evaluated across nine encoders.
- **A comparison of concept supervision:** human supervision consistently outperforms embedding-similarity supervision, highlighting the importance of how concept scores are obtained.
- **An audit of automatic concept scoring:** prompted GPT models are evaluated against the same human concept reference, revealing the distinction between task prediction and concept recovery.

2 Related Work

Concept bottleneck models, from vision to text. CBMs route a prediction through a layer of human-defined concepts and support test-time intervention on those concepts (Koh et al., 2020). Because dense concept annotation is costly, later work replaced it with automatic sources: post-hoc CBMs transfer concepts from other datasets or from natural-language descriptions via multimodal models (Yuksekgonul et al., 2023), LaBo prompts GPT-3 to propose the concept set (Yang et al., 2023), and label-free CBMs avoid concept labels altogether (Oikarinen et al., 2023). Textual CBMs follow the same trajectory. Text Bottleneck Models (Ludan et al., 2024), C^3M (Tan et al., 2025), CB-LLM (Sun et al., 2025), CT-CBM (Bhan et al., 2025) and SCBM (Labadie-Tamayo et al., 2025) obtain concept labels largely through language models, embedding similarity or unsupervised discovery, and are evaluated mainly on English. We keep the bottleneck itself standard and instead vary where the concept scores come from, and we compare these sources against exhaustive human scoring on a low-resource language.

Are the concepts trustworthy? A bottleneck is only interpretable if its concepts mean what their names say. Soft concept probabilities can carry label information beyond the concepts themselves (Mahinpei et al., 2021; Havasi et al., 2022), and concepts predicted by a CBM may not correspond to anything semantically meaningful in the input (Margeloiu et al., 2021). Models trained on randomised concept labels fail to learn the concepts (Ismail et al., 2024), and assessing the quality of a concept set remains an open problem (Galliamov

et al., 2026). These studies examine the *predictor* or the *concept set*. Our question is upstream: given a fixed concept set, how well do automatic *scores* agree with human judgement, and what does it cost downstream when they do not? To our knowledge, this has not been tested with dense human annotation on a pragmatic task.

Sarcasm detection and interpretability. Sarcasm is commonly framed as an incongruity between literal and intended meaning, and surveys trace the move from pattern extraction and hashtag supervision to context beyond the target text (Joshi et al., 2017). Interpretable sarcasm models exist but are few: prototype-based networks explain predictions through similar examples (Wen and Rezapour, 2026), and LLM agents decompose literal meaning, context and intention into interpretable signals (Inoshita and Mizuno, 2025). Neither exposes a vocabulary of human-readable concepts that a user can inspect and score, which is what a bottleneck provides.

Bangla sarcasm. Ben-Sarc supplies 25,636 Bangla Facebook comments for sarcasm detection and reports 75.05% accuracy with a fine-tuned Bangla transformer (Lora et al., 2025); another study reaches 77.1% on it with a Bangla BERT-based generative adversarial model (Lora et al., 2023). Bangla remains under-resourced despite its large speaker population (Joshi et al., 2020). Existing Bangla systems predict a label directly and offer no account of why. We add the first concept-level view of Bangla sarcasm: a concept vocabulary derived from the data and a human-scored matrix of 436,966 binary judgments, which also serves as a reference for auditing automatic scoring.

3 Method

Overview. Figure 2 summarizes the pipeline. From Ben-Sarc, we construct a fixed vocabulary of $K = 43$ concepts and obtain concept scores through human annotation, embedding similarity, and prompted LLMs. For each of nine backbone encoders, we train a direct classifier, a human-supervised CBM, and a similarity-supervised CBM, yielding 27 models under a common train/test split. LLM scores are used only for independent test-set auditing.

3.1 Data

We use 10,162 Ben-Sarc comments (Lora et al., 2025) after removing ambiguous and duplicate entries. A stratified 80/20 split yields 8,129 training comments (4,082 sarcastic; 4,047 non-sarcastic) and 2,033 test comments (1,021 sarcastic; 1,012 non-sarcastic). No re-sampling is applied. For CBM fitting, 6,503 training comments are used for optimization and 1,626 disjoint comments monitor Stage-1 loss; no checkpoint selection or hyperparameter tuning is performed. The test set is reserved for final evaluation.

3.2 Concept Discovery

The 43-concept vocabulary $\mathcal{C}$ is constructed using only the training split before model training and is then fixed across all experiments. Validation and test comments are excluded from concept discovery and vocabulary construction. The resulting vocabulary is applied unchanged to the held-out splits, and no CBM parameters are fitted using test data.

Clustering and labelling. Training comments are clustered separately by class. BanglaBERT embeddings (Bhattacharjee et al., 2022) are reduced from 768 to 50 dimensions using UMAP (McInnes et al., 2018) with cosine distance and clustered using HDBSCAN (McInnes et al., 2017) (Euclidean distance; minimum cluster size 5), producing 78 non-sarcastic and 72 sarcastic clusters. Claude (Anthropic) generates short Bangla descriptors for each cluster.

Consolidation. Descriptors are re-embedded with BanglaBERT and grouped within each class using agglomerative clustering with cosine-distance threshold 0.5, producing 95 non-sarcastic and 96 sarcastic groups. Residual duplicates are merged with Claude Sonnet 4.6 and manually reviewed, yielding 43 concepts: 25 sarcastic-specific and 18 non-sarcastic-specific.

3.3 Concept Supervision

Each comment is scored against all 43 concepts using three sources.

Human annotation. Three annotators worked on the full dataset over a four-month period, labeling 436,966 comment–concept pairs according to predefined annotation guidelines. Each label indicated whether a concept captured the comment's communicative intent or thematic content. Positive labels accounted for 3.3% of test cells (1.42 concepts per comment), while approximately 16% of training comments had no positive concepts. Inter-annotator agreement was assessed using pairwise

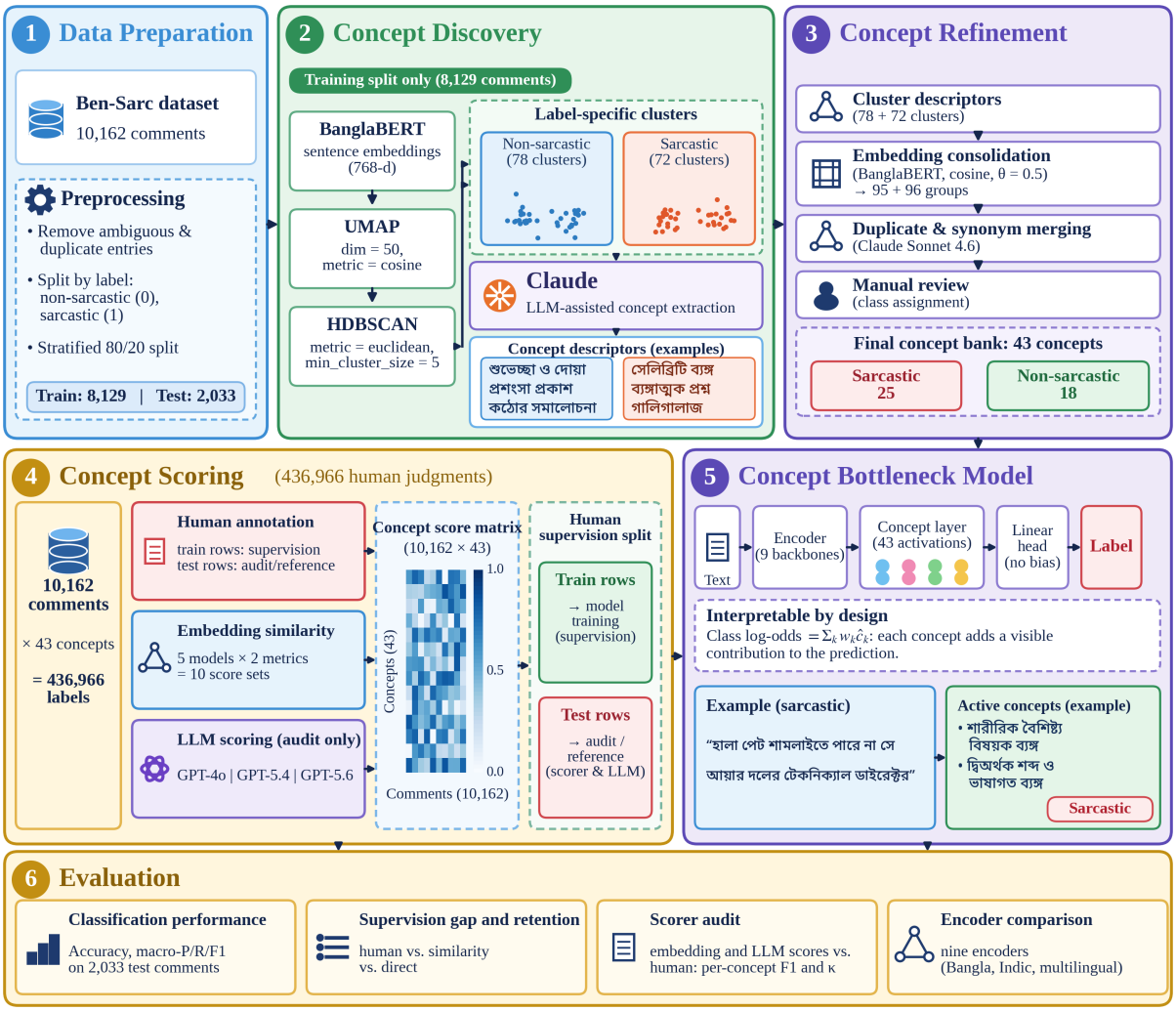


Figure 2: Experimental pipeline. A fixed 43-concept vocabulary is scored by human annotation and embedding similarity to supervise concept bottleneck models, while prompted LLM scorers are audited against the human reference.

Cohen's κ, yielding a mean score of 0.83 across the three annotator pairs.

Embedding similarity. Following CB-LLM (Sun et al., 2025), comment and concept embeddings are compared using cosine similarity or inverted Euclidean distance: The cosine and inverted-Euclidean scores are defined as $\tilde{s}_{ik}^{\cos} = \cos(g(x_i), g(c_k))$ and $\tilde{s}_{ik}^{\text{euc}} = [1 + \|g(x_i) - g(c_k)\|_2]^{-1}$. Five sentence encoders (BanglaBERT, a Bangla sentence transformer, paraphrase-multilingual-MPNet, paraphrase-multilingual-MiniLM, and TigerLLM-1B-it; Raihan and Zampieri, 2025) yield ten candidate score sets across the two measures. The experimentally selected score set is used consistently across all similarity-supervised CBMs.

LLM scoring. GPT-4o (`gpt-4o-2024-11-20`), GPT-5.4 (`gpt-5.4-2026-03-05`), and GPT-5.6 (`gpt-5.6-luna`) score the fixed vocabulary in batches of 20 comments, returning both a sarcasm label and applicable concepts. Their concept assignments and labels are evaluated only on the test set and are never used for CBM training.

3.4 Concept Bottleneck Model

Following Koh et al. (2020) and Sun et al. (2025), each backbone encoder maps a comment to $\mathbf{h}_i \in \mathbb{R}^{768}$ using its `[CLS]` representation, or mean pooling for mT5. A concept layer produces 43 activations followed by a bias-free classification head:

$$\hat{\mathbf{c}}_i = \phi(W_c \operatorname{drop}(\mathbf{h}_i) + \mathbf{b}_c), \qquad \mathbf{z}_i = W_y \hat{\mathbf{c}}_i. \tag{1}$$

Dropout is $p = 0.1$; ReLU is used for human supervision and tanh/sigmoid for similarity supervision

according to score range. The bias-free head makes the prediction a weighted combination of named concept activations.

3.5 Training

CBM training has two stages. First, the encoder and concept layer are aligned with the selected supervision:

Stage 1: concept alignment. The encoder and concept layer are trained to reproduce the selected concept supervision. For human supervision, we minimize cosine distance; for similarity supervision, we minimize mean squared error:

$$\mathcal{L}_{\text{hum}} = \frac{1}{N_{\text{tr}}} \sum_i \left[1 - \cos(\hat{\mathbf{c}}_i, \mathbf{s}_i)\right]. \quad (2)$$

$$\mathcal{L}_{\text{sim}} = \frac{1}{N_{\text{tr}}} \sum_i \|\hat{\mathbf{c}}_i - \tilde{\mathbf{s}}_i\|_2^2. \quad (3)$$

Both CBM settings use AdamW (Loshchilov and Hutter, 2019) with learning rate 2×10^{-5}, weight decay 0.01, cosine annealing, gradient clipping at 1.0, batch size 16, maximum sequence length 128, and 25 epochs. The human objective is scale-invariant and therefore aligns concept directions rather than directly calibrating activations to $\{0, 1\}$.

Second, the encoder and concept layer are frozen and the bias-free head is trained with cross-entropy and L_1 regularization:

$$\min_{W_y} \frac{1}{N_{\text{tr}}} \sum_i \text{CE}(\text{softmax}(W_y \hat{\mathbf{c}}_i), y_i) + \lambda \|W_y\|_1. \quad (4)$$

The head is optimized with Adam (learning rate 10^{-3}) for 25 epochs, with $\lambda = 10^{-3}$ for L_1 regularization. For the BanglaBERT human-supervised CBM, 38/43 concepts remain active under a 10^{-3} weight threshold; hence the head is described as L_1-regularised rather than sparse. The direct baseline fine-tunes the same backbones with a standard classification head and cross-entropy. The nine backbones are multilingual BERT (Devlin et al., 2019), BanglaBERT, a Bangla sentence transformer, XLM-R (Conneau et al., 2020), MuRIL (Khanuja et al., 2021), two IndicBERTv2 variants (Doddapaneni et al., 2023), paraphrase-multilingual-MPNet (Reimers and Gurevych, 2019), and mT5 (Xue et al., 2021).

3.6 Evaluation

All 27 models are evaluated on the same 2,033 test comments using accuracy and macro-averaged precision, recall, and F1; each model is trained once with seed 42. We compare the three classification settings per backbone and report the human–similarity performance gap and human-CBM performance relative to direct classification.

For concept alignment, we report mean Pearson correlation between predicted and human concept scores over concepts with at least one positive human annotation and non-constant predictions, together with excluded-concept counts. We additionally report per-concept F1 after within-split min–max normalization and thresholding at 0.5. F1 is preferred to accuracy because positive concept annotations constitute only 3.3% of test cells.

For each GPT model, concept assignments are compared with the human matrix using macro- and micro-averaged precision, recall, and F1 across the 43 concepts, together with accuracy of the directly predicted sarcasm label.

3.7 Implementation

All models are implemented in PyTorch using Hugging Face Transformers and trained on Google Colab GPUs with random seed 42.

4 Results and Analysis

4.1 Main results

Table 1 and Figure 4a report all 27 models. Human-supervised CBMs outperform similarity-supervised CBMs on every encoder, by 5.8 to 11.9 macro-F1 points (mean +8.6 over the eight encoders on which direct classification works). They also stay close to direct classification: on these eight encoders the human-supervised bottleneck retains 94.8–99.3% of direct accuracy (mean 97.4%), a loss of about two points, whereas the similarity-supervised bottleneck retains 85.8% on average. The best human-supervised CBM (IndicBERTv2-MLM-Sam-TLM, 0.76) is within one point of the best direct classifier (IndicBERTv2-MLM, 0.769). Human-CBM values are available at two decimals, so differences of a point or less between encoders should not be over-interpreted.

4.2 Why similarity supervision falls short

Three observations point to the supervision signal, not the encoder, as the limiting factor. First, the ranges do not overlap: every similarity-supervised CBM (0.609–0.651 macro-F1) is below every human-supervised CBM (0.70–0.76) and every working direct classifier (0.715–0.769). Second,

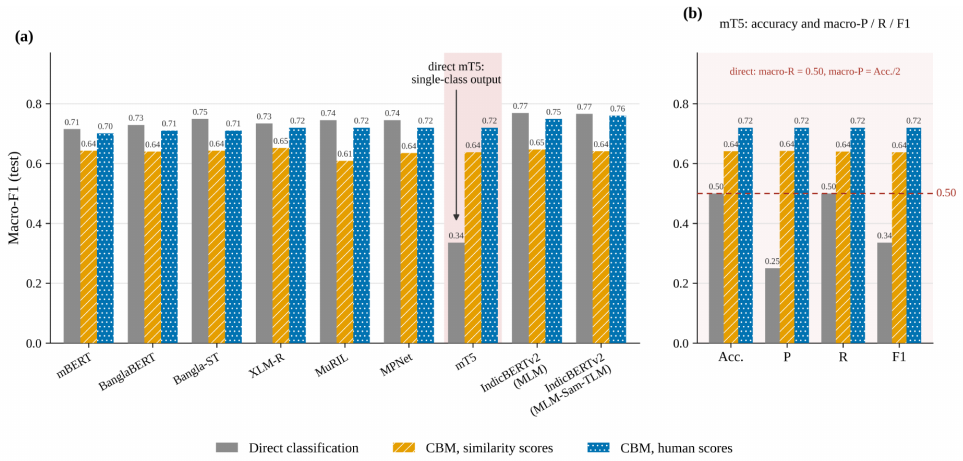


Figure 4: Main results. (a) Macro-F1 of direct classification and of CBMs trained on similarity or human concept scores. (b) For mT5, direct classification outputs a single class (macro-recall 0.50, macro-precision equal to half the accuracy), while both bottlenecks behave like those of the other encoders.

	Direct		CBM, similarity		CBM, human			
Encoder	Acc.	F1	Acc.	F1	Acc.	F1	Δ_e	ρ_e (%)
mBERT	0.716	0.715	0.642	0.642	0.70	0.70	+5.8	97.8
BanglaBERT	0.730	0.729	0.641	0.639	0.71	0.71	+7.1	97.3
Bangla-ST	0.749	0.749	0.642	0.642	0.71	0.71	+6.8	94.8
XLM-R	0.735	0.734	0.651	0.651	0.73	0.72	+6.9	99.3
MuRIL	0.746	0.744	0.609	0.609	0.72	0.72	+11.1	96.5
MPNet	0.744	0.744	0.635	0.635	0.72	0.72	+8.5	96.7
mT5	0.502	0.334	0.639	0.637	0.72	0.72	+8.3	–
IndicBERTv2 (MLM)	0.770	0.769	0.647	0.647	0.75	0.75	+10.3	97.4
IndicBERTv2 (MLM-Sam-TLM)	0.765	0.764	0.641	0.641	0.76	0.76	+11.9	99.3
Mean (8, excl. mT5)	0.744	0.743	0.639	0.638	0.725	0.724	+8.6	97.4

Table 1: Test results for the nine encoders (F1: macro-F1). Δ_e: macro-F1 of the human-supervised minus the similarity-supervised CBM, in points. ρ_e: accuracy of the human-supervised CBM as a percentage of direct accuracy; undefined for mT5, whose direct classifier outputs one class. Each cell is one run (seed 42).

similarity supervision does not transmit encoder quality. The ranking of the eight encoders under direct classification correlates with their ranking under human supervision (Spearman $\rho_s = 0.76$) but hardly with their ranking under similarity supervision ($\rho_s = 0.17$); the encoder that is best under direct classification is not best under similarity supervision, and XLM-R, sixth of eight under direct classification, is first. Third, the structure of the vocabulary makes topical similarity a poor guide to the label. Sixteen of the 43 concepts form eight topic-matched pairs, each of a sarcastic-specific and a non-sarcastic-specific concept (sports, religion, politics, education, health, economy, business, and international affairs; for example, sports sarcasm and sports discussion). An embedding measures topical proximity, so a comment about cricket is about equally close to both members of a pair whether it is sincere or mocking, and its score vector then carries little of the distinction that the label depends on. Human annotators score communicative intent and can separate the pair. The third point is an explanation consistent with the first two rather than a measured effect; we did not test it directly. Because the two pipelines also differ in alignment loss and activation (Section 3.5), part of the gap may come from the training recipe, and we therefore describe it as a difference between supervision pipelines as configured.

4.3 The mT5 collapse

Under direct classification, mT5 obtains 0.502 accuracy and 0.334 macro-F1 (Table 1). Its macro-recall is exactly 0.500 and its macro-precision (0.251) equals half its accuracy, the signature of a model that assigns every comment to one class;

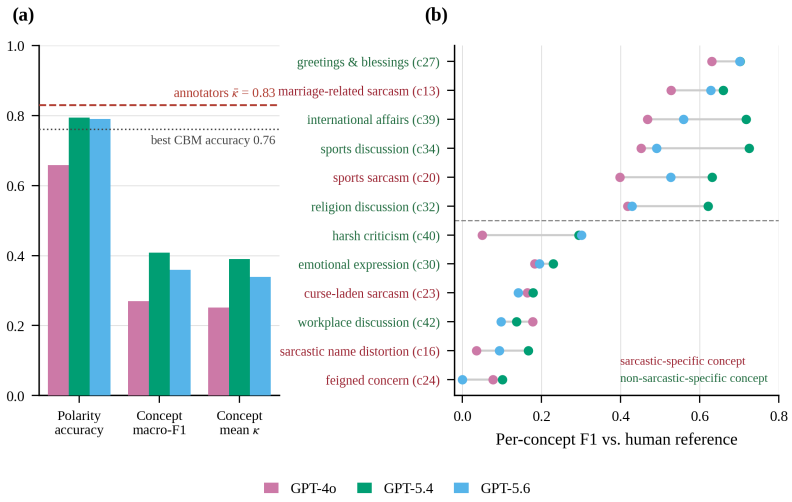


Figure 3: LLM concept scorers vs. the human reference (test split). (a) High polarity accuracy but low concept-level agreement; the dashed line marks annotator agreement ($\bar{\kappa} = 0.83$) and the dotted line the best CBM accuracy (0.76). (b) Per-concept F1 of the three scorers for the six highest- and six lowest-agreement concepts; marker colours denote the scorer, label colours the concept group (red: sarcastic-specific, green: non-sarcastic-specific).

the accuracy of 0.502 corresponds to 1,021 of the 2,033 test comments. This is therefore a collapse to a constant output rather than weak discrimination. The collapse is not a lack of information in the representation. With the same mean-pooled encoder states, a concept layer reaches 0.637 macro-F1 under similarity supervision and 0.72 under human supervision, inside the ranges of the other encoders (Figure 4b). The signal is present in the representation; what fails is fitting it through the direct head. We attribute the failure to the combination of the encoder–decoder pretraining and a weak training signal. mT5 has no `[CLS]` token, so its mean-pooled encoder states are not organised for sequence-level classification, and a single binary label provides a gradient that is easily satisfied by the majority output when the learning rate is small (2×10^{-5}), training lasts at most 25 epochs. A 43-dimensional concept target gives each comment a much richer signal. That even the similarity-supervised bottleneck avoids the collapse shows that the rescue comes from dense concept supervision, not from the human labels as such; the human labels additionally lift performance to the level of the other encoders. We did not ablate pooling, learning rate or seeds, so we read the collapse as a failure of this recipe, not as a general property of mT5.

4.4 Auditing LLM Concept Scorers

Table 2 and Figure 3 compare three prompted LLMs with human annotations on 2,033 test comments. GPT-5.4 and GPT-5.6 achieve polarity accuracies of 79.4% and 79.0%, respectively, exceeding all trained CBMs (best: 0.76) and direct classifiers (best: 0.770); GPT-4o achieves 65.9%. However, concept macro-F1 scores are only 0.41, 0.36, and 0.27, with mean per-concept Cohen's κ scores of 0.39, 0.34, and 0.25, respectively, compared with 0.83 among human annotators. Only 3.3% of concept cells are positive, making accuracy misleading: an all-negative scorer would achieve 96.7%, exceeding the models' 95.5–96.2%. Agreement is higher for explicit topical concepts, such as greetings and blessings (F1: 0.63–0.70), but lower for pragmatic concepts, including feigned concern (0.00–0.10), sarcastic name distortion (0.04–0.17), and curse-laden sarcasm (0.14–0.18). Per-concept rankings remain stable across models (Spearman's ρ: 0.76–0.91), suggesting concept-dependent difficulty. The models also show distinct labeling biases. GPT-5.6 over-predicts sarcastic-specific concepts (2,503 vs. 1,709 human labels; +46%) and under-predicts non-sarcastic-specific concepts (888 vs. 1,173). GPT-4o under-predicts both categories (1,392 and 723), whereas GPT-5.4 most closely matches human counts (1,793 and 1,174). Thus, accurate final-label prediction does not necessarily imply human-aligned concept attribution.

Scorer	Pol. acc.	Concept (macro) P	R	F1	Mean κ	Pred./ gold
GPT-4o	0.659	0.355	0.260	0.270	0.25	0.73
GPT-5.4	0.794	0.453	0.439	0.408	0.39	1.03
GPT-5.6	0.790	0.405	0.407	0.359	0.34	1.18

Table 2: Audit of prompted LLMs against the human matrix on the test split. Pol. acc.: accuracy of the overall sarcastic/non-sarcastic decision. Pred./gold: positive concept decisions relative to the human matrix. Annotator agreement is $\bar{\kappa} = 0.83$.

5 Conclusion

We investigated whether automatically derived concept scores provide the same useful bottleneck representations as dense human supervision for interpretable text classification. Using Bangla sarcasm detection as a test bed, we constructed a fixed vocabulary of 43 concepts and compared human supervision, embedding-similarity supervision, and prompted LLM scoring across nine encoder backbones. Human-supervised CBMs outperform similarity-supervised CBMs across all nine encoders by 5.8–11.9 macro-F1 points, while retaining 95–99% of direct-classification accuracy on the eight encoders with functioning direct classifiers. These results show that concept-based interpretability can approach competitive classification performance without relying on automatically derived concept scores. Our LLM audit reveals a complementary gap: GPT-4o, GPT-5.4, and GPT-5.6 achieve stronger sarcasm-label accuracy than their agreement with the human concept matrix would suggest. Correct final predictions therefore do not necessarily imply accurate human-aligned concept attribution. The mT5 results further illustrate the potential value of concept supervision. Although its direct classifier collapses to a single-class prediction (macro-F1: 0.334), the human-supervised CBM reaches 0.72 accuracy. This finding is specific to the evaluated training configuration and does not establish a general advantage for mT5. Overall, the findings identify supervision source as an important design choice for textual concept bottlenecks: automated scoring reduces annotation requirements but may not capture the pragmatic evidence represented by human annotations. The resulting concept matrix also supports future research on Bangla and other low-resource settings, including studies across languages, tasks, concept granularities, and controlled supervision objectives to identify when human annotation is necessary.

6 Limitations

- **Dataset and task scope.** This study covers Bangla sarcasm detection on a single dataset. Although nine encoders are evaluated, the findings may not generalize to other languages, datasets or tasks. Future work should test the supervision comparison on additional low-resource languages and pragmatic tasks.
- **Concept representation.** Human concept supervision is binary and captures whether a concept applies, not its degree of relevance. Future work could investigate graded annotations, concept uncertainty and relationships between concepts.
- **Supervision and training configuration.** Human- and similarity-supervised CBMs use different alignment objectives and activations. The observed gap thus reflects the complete supervision pipelines, not an isolated causal effect of the supervision source. Controlled experiments varying only the concept scores could provide a more direct comparison.
- **Evaluation scale.** Each configuration uses a single random seed on one task and language, so we report observed differences, not statistical significance. Multiple seeds and additional datasets would give stronger evidence of robustness.
- **Reproducibility.** We plan to release the dataset, annotation guidelines, and experimental code upon publication to facilitate independent verification and future research.

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A Additional Encoder-Level Results

This section complements Table 1 and the main results of the paper with a per-encoder view of the same eight runs. We leave out mT5, whose direct classifier collapses to a single class (Section 4.3), so the retention ratio below is undefined for it. All numbers are taken from Table 1 and come from single runs with seed 42.

A.1 Macro-F1 and Accuracy Retention per Encoder

Figure 5a plots, for every encoder, the macro-F1 under direct classification, the human-supervised CBM and the similarity-supervised CBM. Moving from direct classification to the human-supervised bottleneck lowers macro-F1 only slightly. The drop is between 0.4 and 2.4 points for seven encoders and about 3.9 points for Bangla-ST ($0.749 \rightarrow 0.71$). Moving to the similarity-supervised bottleneck lowers it much more, to 0.609–0.651 for all eight encoders, and the steepest fall is that of MuRIL ($0.744 \rightarrow 0.609$). Human-CBM scores are available at two decimals only, so differences of about one point should not be over-interpreted.

The inset of Figure 5a gives the Spearman correlation between the ranking of the eight encoders under direct classification and their ranking under each bottleneck: $\rho_s = 0.76$ for human supervision and $\rho_s = 0.17$ for similarity supervision. Under human supervision, an encoder that is strong as a direct classifier therefore tends to remain strong; under similarity supervision this relationship largely disappears. For example, XLM-R is sixth of eight under direct classification but first under similarity supervision. This supports the observation in Section 4.2 that similarity supervision does not transmit encoder quality.

Figure 5b shows the retention ratio, i.e. the accuracy of the CBM divided by the accuracy of the direct classifier of the same encoder (the quantity ρ_e in Table 1). The human-supervised CBM retains 94.8–99.3% of direct accuracy (mean 97.4%), which we summarise as roughly 95–99% in the main text. The similarity-supervised CBM retains 81.6–89.7% (mean 85.8%). For every encoder the human-supervised CBM retains more than the similarity-supervised CBM, by 8 to 16 percentage points, and the two ranges do not overlap. The best human-supervised retention is reached by XLM-R and IndicBERTv2 (MLM-Sam-TLM), both at 99.3%, and the lowest by Bangla-ST at 94.8%. As noted in Section 4.2, the two pipelines also differ in their alignment loss and activation function, so the figure compares the supervision pipelines as configured, not the supervision source alone.

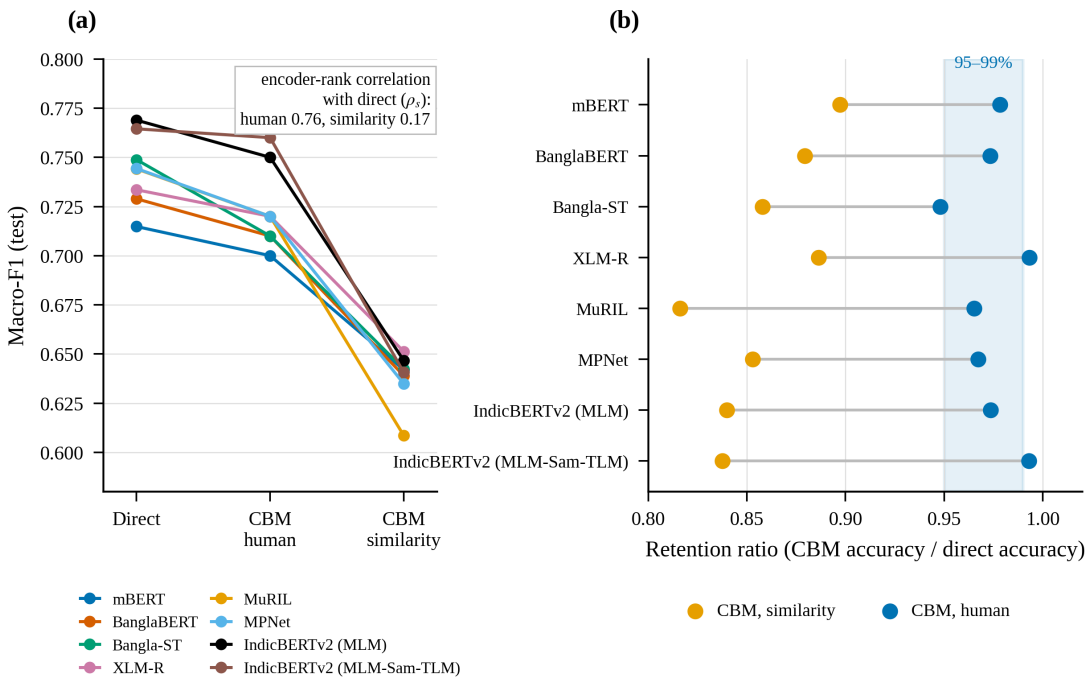


Figure 5: Encoder-level test results for eight encoders (mT5 is omitted because its direct classifier predicts a single class). (a) Macro-F1 for direct classification, human-supervised CBM, and similarity-supervised CBM; the inset shows Spearman's ρ_s between direct-classifier rankings and CBM rankings. (b) Retention ratio, defined as CBM accuracy divided by direct-classifier accuracy, for similarity-supervised and human-supervised CBMs. The shaded band indicates the 95–99% range reported in the text.

B Concept-Neuron Visualisations

This appendix inspects the concept layer of the human-supervised CBM. Each of the K=43 concept neurons corresponds to one named concept, and the bias-free head W_y turns the activations $\hat{c}_i$ into the class logits. We show (i) how neurons activate on average for each class (Section B.1) and (ii) which neurons carry the most weight for each class (Section B.2). Section B.3 describes how the figures were produced. Concept names are in Bangla inside the figures; we give English glosses in the text.

B.1 Class-Level Activation Patterns

Figure 6 shows the mean activation of every concept neuron over the sarcastic (left) and non-sarcastic (right) comments. Each outer node is one concept neuron; its colour is the mean activation and its size is the absolute difference between the two class weights of that neuron in the linear head. Two patterns stand out. First, the neurons of the class that matches the text are clearly more active than those of the other class: on sarcastic comments, the non-sarcastic concept neurons stay uniformly pale, and on non-sarcastic comments the sarcastic concept neurons are mostly weak. Second, activation is concentrated on a few concepts rather than spread evenly. Among the sarcastic comments, a handful of nodes are clearly darker than the rest; among the non-sarcastic comments, the sports-discussion neuron is the strongest, followed by the neurons for praise, greetings and wishes, and curiosity. Some sarcastic concept neurons also reach moderate activation on non-sarcastic comments (right panel), so the sarcastic concepts are not perfectly class-specific; this is consistent with the topic overlap between sarcastic and non-sarcastic concepts discussed in the main text.

B.2 Concept Neurons with the Largest Class Weights

The bias-free head makes each logit a weighted sum of concept activations, so the weights W_y show which neurons the model relies on most for each class. Figure 7 lists the five concepts with the largest weight for each class. For the sarcastic class they are television-show mockery (w=0.84), political sarcasm (0.78), sports sarcasm (0.66), education sarcasm (0.61) and double-meaning words and linguistic sarcasm (0.59). For the non-sarcastic

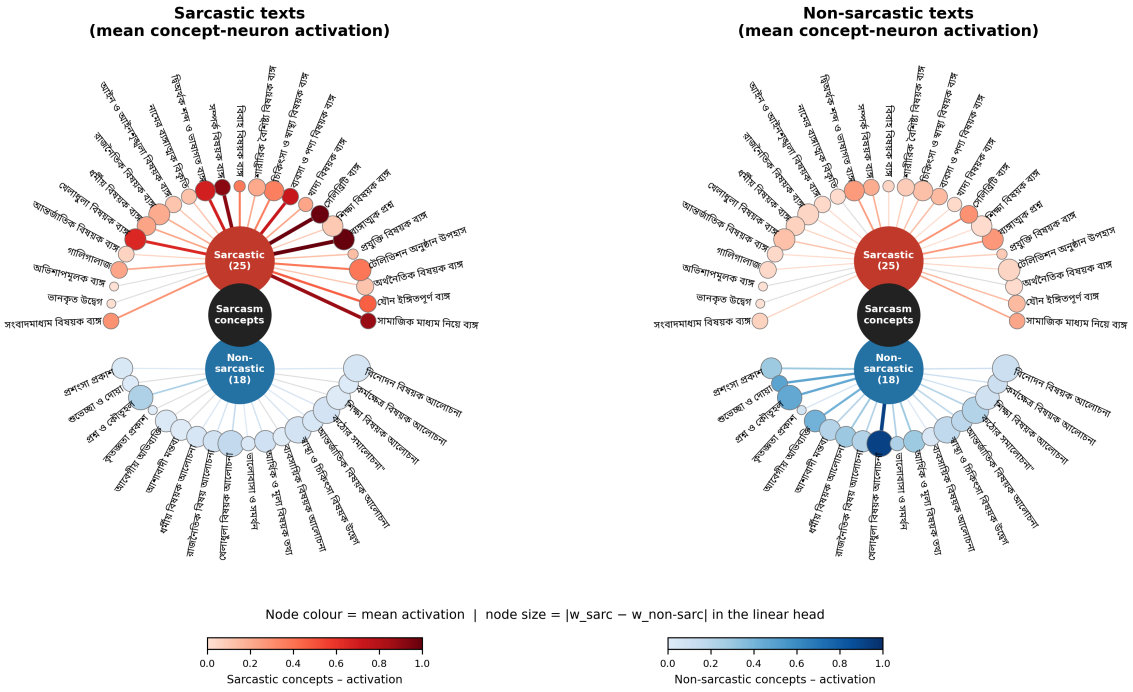


Figure 6: Mean concept-neuron activation on sarcastic (left) and non-sarcastic (right) comments. The upper arc shows the 25 sarcastic-specific concepts and the lower arc the 18 non-sarcastic-specific concepts. Node colour is the mean activation; node size is $|w_{\text{sarc}} - w_{\text{non}}|$ in the linear head; edge colour and width follow the activation.

class they are sports discussion (1.19), entertainment discussion (1.12), health and medical concern (1.10), harsh criticism (1.09) and question and curiosity (0.96). Two observations follow. The top sarcastic weights are dominated by topic-specific sarcasm concepts, and only one of the five (double-meaning words) is a purely linguistic device. The non-sarcastic weights are larger in magnitude, and sports appears in both lists (sports sarcasm versus sports discussion), which matches the topic-matched concept pairs described in the main text: the head must separate two concepts about the same topic, which is where human-scored, intent-based concepts are expected to help over topical similarity. We read the weights as evidence about the trained head, not as a causal attribution of individual predictions.

B.3 Details of the Visualisations

Activations. For a comment x_i, the activation of concept neuron k is $\hat{c}_{ik}$, the output of the concept layer of the human-supervised CBM. The class-level view in Figure 6 averages $\hat{c}_{ik}$ over the comments of each class and shows the values on a common $[0, 1]$ colour scale.

Class weights. Figure 7 and the node sizes in Figure 6 use the weights of the bias-free classification head W_y (size 2×43) after the second training stage.

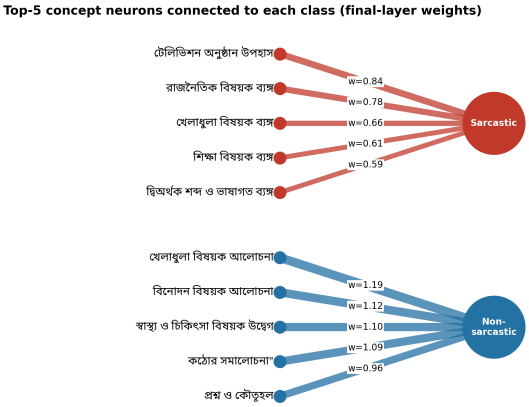


Figure 7: The five concept neurons with the largest weight in the final layer for the sarcastic class (top) and the non-sarcastic class (bottom). Edge width and the label w show the weight.